

# Certainty Equivalent: Simple Explanation

## What is a Certainty Equivalent?

### Simple Definition:

The certainty equivalent (CE) is the **guaranteed amount of money** that makes you exactly as happy as taking a risky gamble.

In other words: "How much would I need for certain to make me indifferent to this lottery?"

## The Formula

The CE is defined by:

$$u(CE) = EU$$

where:

- $u$  = your utility function
- $CE$  = certainty equivalent (what we're solving for)
- $EU$  = expected utility of the gamble

**In words:** The utility of getting CE for certain equals the expected utility of the gamble.

## Simple Example

You're offered a coin flip:

- Heads: win \$100
- Tails: win \$0

If you're risk-neutral, your  $CE = \$50$  (the expected value).

But if you're risk-averse, you'd say: "I'd be just as happy with \$40 guaranteed as I would be with this 50-50 gamble for \$100." Then your  $CE = \$40$ .

**Key insight:** The more risk-averse you are, the LOWER your certainty equivalent!

# CRRA Utility Example

## The Setup

Let's say you use CRRA utility:

$$u(x) = \frac{x^{1-\sigma}}{1-\sigma}$$

where  $\sigma$  is your risk aversion parameter (higher  $\sigma$  = more risk averse).

## The Gamble

50-50 chance between:

- \$0 with probability  $\frac{1}{2}$
- \$200 with probability  $\frac{1}{2}$

## Finding the Certainty Equivalent

### Step 1: Calculate Expected Utility

$$EU = \frac{1}{2} \cdot u(0) + \frac{1}{2} \cdot u(200)$$

With CRRA utility:

$$EU = \frac{1}{2} \cdot \frac{0^{1-\sigma}}{1-\sigma} + \frac{1}{2} \cdot \frac{200^{1-\sigma}}{1-\sigma}$$

Since  $0^{1-\sigma} = 0$ :

$$EU = \frac{1}{2} \cdot \frac{200^{1-\sigma}}{1-\sigma}$$

**Step 2: Set  $u(CE) = EU$**

$$\frac{CE^{1-\sigma}}{1-\sigma} = \frac{1}{2} \cdot \frac{200^{1-\sigma}}{1-\sigma}$$

**Step 3: Cancel out the  $(1-\sigma)$  terms**

$$CE^{1-\sigma} = \frac{1}{2} \cdot 200^{1-\sigma}$$

**Step 4: Solve for CE**

Take both sides to the power of  $\frac{1}{1-\sigma}$ :

$$CE = \left(\frac{1}{2}\right)^{\frac{1}{1-\sigma}} \cdot 200$$

**The Answer:**

$$CE = \left(\frac{1}{2}\right)^{\frac{1}{1-\sigma}} \cdot 200$$

## What Does This Mean?

Let's plug in different values of  $\sigma$  (risk aversion):

### Case 1: Risk Neutral ( $\sigma = 0$ )

$$CE = \left(\frac{1}{2}\right)^{\frac{1}{1-0}} \cdot 200 = \left(\frac{1}{2}\right)^1 \cdot 200 = \frac{1}{2} \cdot 200 = 100$$

Makes sense! If you're risk-neutral, you only care about expected value:

$$EV = \frac{1}{2}(0) + \frac{1}{2}(200) = 100$$

### Case 2: Slightly Risk Averse ( $\sigma = 0.5$ )

$$CE = \left(\frac{1}{2}\right)^{\frac{1}{1-0.5}} \cdot 200 = \left(\frac{1}{2}\right)^2 \cdot 200 = \frac{1}{4} \cdot 200 = 50$$

You'd need \$50 guaranteed to be as happy as the gamble.

### Case 3: More Risk Averse ( $\sigma = 2$ )

$$CE = \left(\frac{1}{2}\right)^{\frac{1}{1-2}} \cdot 200 = \left(\frac{1}{2}\right)^{-1} \cdot 200 = 2 \cdot 200 = 400$$

Wait, that's weird! This says  $CE = \$400$ , but the most you can win is \$200!

**What's happening?** When  $\sigma > 1$ , the exponent  $\frac{1}{1-\sigma}$  becomes negative, which flips the fraction. This is mathematically correct but the interpretation changes.

For  $\sigma > 1$  with this particular gamble (one outcome is \$0), the utility function becomes very sensitive to getting zero, and the math doesn't give us the intuitive CE we expect.

### Case 4: Log Utility ( $\sigma = 1$ )

When  $\sigma = 1$ , we use  $u(x) = \ln(x)$  instead:

$$EU = \frac{1}{2} \ln(0) + \frac{1}{2} \ln(200)$$

But  $\ln(0) = -\infty$ , so this gamble has infinite negative utility! You'd reject it at any price.

#### Important Note:

When one outcome is \$0 and you have log utility or high risk aversion, the gamble can have very low (or even infinitely negative) utility because you really, really don't want to end up with \$0.

That's why in practice, these problems usually give you gambles where all outcomes are positive amounts.

# The Big Picture

## How Risk Aversion Affects CE

| Risk Aversion ( $\sigma$ ) | CE       | Interpretation          |
|----------------------------|----------|-------------------------|
| 0 (risk-neutral)           | \$100    | CE = Expected Value     |
| 0.5 (slightly risk-averse) | \$50     | Need \$50 for sure      |
| Higher $\sigma$            | Lower CE | Need even less for sure |

**The pattern:** As you become MORE risk-averse (higher  $\sigma$ ), your certainty equivalent DECREASES.

**Intuition:** "I really hate risk, so I'll accept a much smaller guaranteed amount instead of gambling."

## Risk Premium

The **risk premium** is how much you're willing to give up to avoid risk:

$$\text{Risk Premium} = E[x] - CE$$

For our example with  $\sigma = 0.5$ :

$$RP = 100 - 50 = 50$$

You'd pay up to \$50 to avoid this risk!

## More Realistic Example

Let's do one where all outcomes are positive:

### Gamble: 50-50 between \$100 and \$200

Expected value:  $E[x] = \frac{1}{2}(100) + \frac{1}{2}(200) = 150$

### Finding CE with CRRA utility

$$EU = \frac{1}{2} \cdot \frac{100^{1-\sigma}}{1-\sigma} + \frac{1}{2} \cdot \frac{200^{1-\sigma}}{1-\sigma}$$

$$\frac{CE^{1-\sigma}}{1-\sigma} = \frac{1}{2} \cdot \frac{100^{1-\sigma}}{1-\sigma} + \frac{1}{2} \cdot \frac{200^{1-\sigma}}{1-\sigma}$$

Cancel  $(1-\sigma)$ :

$$CE^{1-\sigma} = \frac{1}{2} (100^{1-\sigma} + 200^{1-\sigma})$$

Solve for CE:

$$CE = \left[ \frac{1}{2} (100^{1-\sigma} + 200^{1-\sigma}) \right]^{\frac{1}{1-\sigma}}$$

## Numerical Examples

**Risk-neutral** ( $\sigma = 0$ ):

$$CE = \left[ \frac{1}{2}(100 + 200) \right]^1 = \frac{300}{2} = 150$$

CE = Expected Value, as expected!

**Risk-averse** ( $\sigma = 0.5$ ):

$$CE = \left[ \frac{1}{2}(100^{0.5} + 200^{0.5}) \right]^2 = \left[ \frac{1}{2}(10 + 14.14) \right]^2 \approx (12.07)^2 \approx 145.7$$

So you'd accept \$145.70 guaranteed instead of the 50-50 gamble for \$100 or \$200.

Risk premium:  $150 - 145.7 = \$4.30$

# Summary

## Key Takeaways:

1. **Certainty Equivalent (CE):** The guaranteed amount that makes you as happy as taking the gamble

$$u(CE) = EU$$

2. **Finding CE:**

- Calculate expected utility  $EU$
- Solve  $u(CE) = EU$  for  $CE$

3. **Risk aversion and CE:**

- More risk-averse  $\Rightarrow$  Lower CE
- Risk-neutral  $\Rightarrow$  CE = Expected Value

4. **Risk Premium:**

$$RP = E[x] - CE$$

How much you'd pay to avoid risk

5. **CRRA utility:**  $u(x) = \frac{x^{1-\sigma}}{1-\sigma}$

- $\sigma$  = risk aversion parameter
- Higher  $\sigma$  = more risk-averse = lower CE

6. **Watch out:** When outcomes include \$0 and  $\sigma \geq 1$ , utility can be  $-\infty$  (you really hate getting zero!)

## Exam Tips

- **Always start with:**  $u(CE) = EU$
- **For CRRA:** The  $(1 - \sigma)$  denominators will cancel when you set  $u(CE) = EU$
- **Don't forget:** Take the  $\frac{1}{1-\sigma}$  power at the end to solve for CE
- **Sanity check:**
  - CE should be between the minimum and maximum outcomes (usually)
  - $CE \leq$  Expected Value (if risk-averse)
  - $CE =$  Expected Value (if risk-neutral)
- **Special case:** If  $\sigma = 1$ , use log utility:  $u(x) = \ln(x)$